Fundamentals of Computer Systems (COSI 131A)

Lectures: **Tue/Thu 3:55 PM – 5:15 PM** @ **Golding Judaica: 110**
Recitation*: **Wed 5:40 PM – 7:40 PM** @ **Golding Judaica: 110**
Class Website | Gradescope | Moodle

Course Description

This course is an introduction to computer systems organization and operating systems. The objectives of the course are for you to learn three things. The **first thing** is how computer systems, operating systems, and, more generally, computers work. The goal is to demystify the interactions between the software you have written in other courses and hardware. A student graduating with a CS degree should be able to describe the chain of computer system events that occurs from when a user hits the reboot button to when the user's program runs, reading input and displaying results, from the register level to the operating system level to the application level. This is philosophically important, but it is also of practical interest to developers who need to figure out how to make a system do what they want it to do. The **second thing** is for you to learn the fundamental concepts of virtualization, concurrency, and persistence. These ideas are often best explained as abstractions that some software layer (usually the operating system) provides above imperfect hardware to make that hardware usable by programmers and users. The intent is for you to understand such abstractions well enough to be able to synthesize new abstractions when faced with new problems. The ideas and abstractions that we will cover are relevant not only to operating systems but also to many large-scale systems. Thus, a **third goal** of this course is to enhance your ability to understand, design, and implement such systems.

Prerequisites

COSI 12B and **COSI 21A**. A working knowledge of Linux and Java is expected. The programming assignments will be in Java and C and must run on Linux.

Instructor & Teaching Assistant(s)

	Contact Information	**Student Hours**
Subhadeep Sarkar Instructor	Office: Volen 259 Email: subhadeep@brandeis.edu	Tue/Thu: 10:00 AM – 11:00 AM
Shubham Kaushik Teaching assistant	Office: Volen 104 (Vertica Lounge) Email: kaushiks@brandeis.edu	Mon: 2:00 PM – 3:30 PM Fri: 10:00 AM – 11:30 AM

*** Attending the Recitation sessions is key to being successful in the class.**

Class Resources & Contact Policy

The class website is updated in real time with the class. **This is your go-to** for the schedule, readings, and general resources about the class. This is also where all **written and programming assignments will be posted**, along with other relevant resources for the class.

Moodle will be the preferred mode of communication with the instructor and teaching assistants (TAs) outside of class hours and student hours. Posts on Moodle will be responded to within 24 hours. Throughout the course, students are encouraged to ask conceptual questions and start discussions on Moodle. Other students should feel free to answer any questions and participate in any discussions on Moodle. The teaching staff will go through the Moodle questions the same day that student hours are held.

Emails sent during weekdays should expect a response within 48 hours. Be aware that not receiving a response is not a valid reason to not hand in homework on time, so start your work early to make sure you have no questions.

We will be using Gradescope for grading assignments. Please register on Gradescope, using access code: **ZJ7D2R**.

Required Textbook

Operating System Concepts. Abraham Silberschatz, Greg Gagne, and Peter B. Galvin. 10th Edition. Wiley. ISBN: 978-1-119-32091-3. (A 120-day ebook rental is available for around $29.) Referred to as OSC on the schedule.

Additional recommended readings (optional):
Operating Systems: Three Easy Pieces. Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau. Free online at pages.cs.wisc.edu/~remzi/OSTEP. Referred to as OSTEP on the schedule.

Learning Goals

Students who successfully complete all components of this course will be able to demonstrate the following by the end of the semester.

a) Understanding of the hardware abstraction layer and machine organization: how instructions, memory, and I/O are represented and executed at the machine level.
b) Understanding of the operating system's virtualization of the CPU — processes, threads, kernel/user mode, and CPU scheduling policies.
c) Ability to reason about concurrency and write correct synchronized programs using locks, condition variables, and related primitives, and to recognize and avoid deadlock.
d) Understanding of the operating system's virtualization of memory — memory management and virtual memory (paging).
e) Working proficiency in C, including awareness of common memory-safety risks and mitigations, and the ability to translate higher-level (Java) designs into C.

f) Understanding of persistence: file systems, disk organization, naming, and crash-consistent recovery/transactions.
g) Familiarity with virtual machines and how they relate to the rest of the systems stack.

Components of Course Work

Success in this four-credit course is based on the expectation that students will spend a minimum of 9 hours of study time per week in preparation for class. As with most technical courses, besides ability and motivation, it takes time to learn and master the subject.

A. *Class Participation and Interaction:* The class is designed to be interactive. The students are highly encouraged to ask questions and participate in in-class discussions. All lectures are delivered in person. The students are expected to attend all lectures and regularly interact with the teaching staff during student hours.
B. *Tutorials:* There will be an introductory tutorial/review for each project. The tutorials are not mandatory but highly recommended. The date and time of the tutorial will be announced in advance in the lecture and also posted on the class website and Moodle close to the tutorial date once they are finalized. Typically, the tutorials run during the recitations.
C. *Programming Assignments:* During the semester, there will be 4-5 programming assignments. This is perhaps the most valuable part of this class. The projects will be based on topics taught in class -- they will require you to apply in practice the concepts covered in the lectures and will require you to defend your design decisions. The projects will be done individually. The projects will focus on concurrency, a concept new to most of you. You will implement a sequential solution to a problem and learn how to turn your sequential solution into a concurrent one using the Java concurrency library. You will gain experience developing a concurrent solution directly using Java default synchronization, and learn how to improve its efficiency using Java general synchronization. Once you have mastered the programming concepts and principles, the choice of a specific programming language becomes less important. To this end, you will learn how to translate your Java solutions safely into C.
D. *Problem Sets:* Periodic written assignments based on concepts taught in class. The problem sets will be graded based on correctness.
E. *Quizzes:* There will be three (alternatively, two) mandatory in-class quizzes testing the mastery of the material covered in the course. Unless stated otherwise, the quizzes will be closed-book and will cover material from lectures, readings, problem sets, and the programming assignments. Your success in the quizzes is subject to (i) **attending the class regularly**, (ii) submitting all assignments on time to receive feedback, and (iii) attending the recitation sessions. There is no final exam.

Group work and collaboration are essential to your learning. That said, all of your problem sets and programming assignments must be your own original work and produced without the assistance of others unless I specifically state otherwise.

Evaluation and Grading

The breakdown of the course grade is as follows (minor alterations may occur).

Class Element	Grade Percentage
In-class participation	*5%*
Problem sets	*15%*
Programming assignments	*40%*
Quizzes	*40%*

Late Policy

Every student is granted a cumulative allowance of 4 late days, serving as extensions for their individual assignments, and no penalties are incurred during this period. These late days can be utilized across various assignments. However, once the allotment of 4 late days is exhausted, a penalty of 20% per day is imposed on overdue homework assignments.

Tentative Schedule

The full lecture-by-lecture schedule with dates and reading assignments is maintained on the class website's Schedule page and updated there as the semester is finalized. Overview by unit:

Week #	Topic
1	Introduction to OS + Assembly programming
2	Hardware Abstraction Layer - I
3	Hardware Abstraction Layer - II
4	Kernel Abstraction + Processes and threads
5	Virtualization: CPU
6	CPU Scheduling
7	Synchronization - I
8	Synchronization - II
9	Virtualization: Memory
10	Programming in C
11	Persistence - I
12	Persistence - II
13	Review & Midterm II

Important Course Policies

Academic honesty

You are expected to be familiar with, and to follow, the University's policies on academic integrity. You are expected to be honest in all of your academic work. Please consult Brandeis University Rights and Responsibilities for all policies and procedures related to academic integrity. Allegations of academic dishonesty will be forwarded to Student Rights and Community Standards. Sanctions for academic dishonesty can include failing grades and/or suspension from the university. Citation and research assistance can be found on the university library website.

Accommodations

Brandeis seeks to create a learning environment that is welcoming and inclusive of all students, and I want to support you in your learning. If you think you may require disability accommodations, you will need to work with Student Accessibility Support (SAS). You can contact them at 781-736-3537, email them at access@brandeis.edu, or visit the Student Accessibility Support home page. You can find helpful student FAQs and other resources on the SAS website, including guidance on how to know whether you might be eligible for support from SAS.

If you already have an accommodation letter from SAS, please provide me with a copy as soon as you can so that I can ensure effective implementation of accommodations for this class. In order to coordinate exam accommodations, ideally you should provide the accommodation letter at least 48 hours before an exam.

Respectful environment

Brandeis University is committed to providing its students, faculty, and staff with an environment conducive to learning and working, where all people are treated with respect and dignity. Please refrain from any behavior toward members of our Brandeis community, including students, faculty, staff, and guests, that intimidates, threatens, harasses, or bullies.

Laptop computer and cell phone use

To create a focused and distraction-free learning environment, we have implemented a no laptop/mobile policy during the class. This policy is aimed at maximizing student engagement, fostering active participation, and promoting a conducive atmosphere for effective learning. By minimizing the use of laptops and mobile devices, we aim to enhance the overall quality of the learning experience and encourage direct interaction with course materials and fellow students.

Use of generative AI tools

While there will be portions of the course that do not allow the use of AI tools, such as in-person exams, this course does not prohibit student usage of AI for assistance in completing assignments. If you do make use of AI for completing assignments, you must copy your prompts and submit them to an internal course tool that will be shared on the Moodle page. Recording your AI interactions during assignment completion is comparable to citing sources and related work in writing tasks. If you are found to have used an AI assistant to generate answers for your coursework and you do not report that usage, you may be in violation of the university's honor code and may be penalized. In addition, you are encouraged to consider how the use of AI for these activities may impact your performance on the activities where AI is prohibited. Please note that students are not required to use AI at all if they don't want to, and the course won't be designed in a way that requires students to use AI.

Financial aid for purchasing course materials

If you have difficulty purchasing course materials, please make an appointment with your Student Financial Services or Academic Services advisor to discuss possible funding options and alternative solutions.

Student Support

Success in this course depends heavily on your personal health and well-being. Recognize that stress is an expected part of the college experience, and it often can be compounded by unexpected setbacks or life changes outside the classroom. Your other professors and I strongly encourage you to reframe challenges as unavoidable pathways to success. Reflect on your role in taking care of yourself throughout the academic year, before the demands of exams and projects reach their peak. Please feel free to reach out to me about difficulties you may be having that may impact your performance in this course as soon as it occurs and before it becomes too overwhelming.